$$P_{01} \times Q_{01} = \frac{\sum p_1 q_1}{\sum p_0 q_0}$$

### Example 9.14

Construct Fisher's price index number and prove that it satisfies both Time Reversal Test and Factor Reversal Test for data following data.

| Commodities | Base Year | | Current Year | |
|---|---|---|---|---|
| | Price | Quantity | Price | Quantity |
| Rice | 40 | 5 | 48 | 4 |
| Wheat | 45 | 2 | 42 | 3 |
| Rent | 90 | 4 | 95 | 6 |
| Fuel | 85 | 3 | 80 | 2 |
| Transport | 50 | 5 | 65 | 8 |
| Miscellaneous | 65 | 1 | 72 | 3 |

***Solution:***

| Commodities | Base Year | | Current Year | | $p_0q_0$ | $p_0q_1$ | $p_1q_0$ | $p_1q_1$ |
|---|---|---|---|---|---|---|---|---|
| | Price ($p_0$) | Quantity ($q_0$) | Price ($p_1$) | Quantity ($q_1$) | | | | |
| Rice | 40 | 5 | 48 | 4 | 200 | 160 | 240 | 192 |
| Wheat | 45 | 2 | 42 | 3 | 90 | 135 | 84 | 126 |
| Rent | 90 | 4 | 95 | 6 | 360 | 540 | 380 | 570 |
| Fuel | 85 | 3 | 80 | 2 | 255 | 170 | 240 | 160 |
| Transport | 50 | 5 | 65 | 8 | 250 | 400 | 325 | 520 |
| Miscellaneous | 65 | 1 | 72 | 3 | 65 | 195 | 72 | 216 |
| | | | | **Total** | 1220 | 1600 | 1341 | 1784 |

Table 9.13

Fisher's price index number

$$P_{01}^{F} = \left( \sqrt{\frac{\sum p_1 q_0 \times \sum p_1 q_1}{\sum p_0 q_0 \times \sum p_0 q_1}} \right) \times 100$$

$$= \left( \sqrt{\frac{1341 \times 1784}{1220 \times 1600}} \right) \times 100 = 110.706$$

Time Reversal Test: $P_{01} \times P_{10} = 1$

$$P_{01} \times P_{10} = \sqrt{\left( \frac{\sum p_1 q_0 \times \sum p_1 q_1 \times \sum p_0 q_1 \times \sum p_0 q_0}{\sum p_0 q_0 \times \sum p_0 q_1 \times \sum p_1 q_1 \times \sum p_1 q_0} \right)}$$

$$P_{01} \times P_{10} = \sqrt{\left( \frac{1341 \times 1784 \times 1600 \times 1220}{1220 \times 1600 \times 1784 \times 1341} \right)}$$

$$P_{01} \times P_{10} = 1$$

Factor Reversal Test

$$P_{01} \times Q_{01} = \frac{\sum p_1 q_1}{\sum p_0 q_0}$$

$$P_{01} \times Q_{01} = \sqrt{\left( \frac{\sum p_1 q_0 \times \sum p_1 q_1 \times \sum q_1 p_0 \times \sum q_1 p_1}{\sum p_0 q_0 \times \sum p_0 q_1 \times \sum q_0 p_0 \times \sum q_0 p_1} \right)}$$

$$P_{01} \times Q_{01} = \sqrt{\left( \frac{1341 \times 1784 \times 1600 \times 1784}{1220 \times 1600 \times 1220 \times 1341} \right)}$$

$$P_{01} \times Q_{01} = \sqrt{\left( \frac{1784 \times 1784}{1220 \times 1220} \right)} = \frac{1784}{1220}$$

$$\Rightarrow \quad P_{01} \times Q_{01} = \frac{\sum p_1 q_1}{\sum p_0 q_0}$$

## 9.2.4 Construction of Cost of Living Index Number

Cost of Living Index Number is constructed to study the effect of changes in the price of goods and services of consumers for a current period as compared with base period. The change in the cost of living index number between any two periods means the change in income which will be necessary to maintain the same standard of living in both the periods. Therefore the cost of living index number measures the average increase in the cost to maintain the same standard of life. Further, the consumption habits of people differ widely from class to class (rich, poor, middle class) and even with the region. The changes in the price level affect the different classes of people, consequently the general price index numbers fail to reflect the effect of changes in their cost of living of different classes of people. Therefore,



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